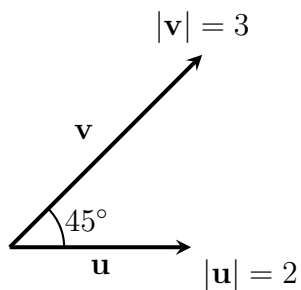

Instructions. (15 points) Sections 12.1–12.4. Show relevant work for full credit.

- (3^{pts}) **1.** Use the figure below to answer the questions that follow. The vectors $\mathbf{u}$ and $\mathbf{v}$ lie in the plane of the page.



- (a) Find $\mathbf{u} \cdot \mathbf{v}$.
- (b) Find $|\mathbf{u} \times \mathbf{v}|$.
- (c) Is $\mathbf{u} \times \mathbf{v}$ directed into the page or out of the page?

- (4^{pts}) **2.** Find two unit vectors that are orthogonal to both $\mathbf{j} + 2\mathbf{k}$ and $\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$.

(4^{pts}) **3.** Find the vector projection of $\mathbf{b} = \langle 12, 1, 2 \rangle$ onto $\mathbf{a} = \langle -1, 4, 8 \rangle$.

(4^{pts}) **4.** Find the angle at vertex B , to the nearest tenth of a degree, for the triangle with vertices $A(1, 0, -1)$, $B(3, -2, 0)$, and $C(1, 3, 3)$.